Kỳ hạn bảo quản biểu ghi chép hiệu chỉnh : Vô thời hạn | Rev 01(2022/06/20)

BIỂU GHI CHÉP KẾT QUẢ HIỆU CHUẨN DỤNG CỤ THIẾT BỊ

Tên dụng cụ thiết bị :	**Tấm độ nhám bề mặt chuẩn**			Mã số quản lý:	**S24**	**01**	**0005**
Nhà sản xuất:	**Accretech**	Phòng ban:	**P.Kiểm tra**	Chu kỳ HC:	12		Tháng
Model:	**E-MC-S24D**	Công đoạn:	**KT-HC**	Ngày hiệu chỉnh:	**2025/12/05**		
Serial NSX:	**60994620LZ**	Chủng loại:	**Dùng chung**	Kỳ hạn hữu hiệu:	**2026/12/31**		
Serial VDM:	**-**	Người đảm trách:	**Thảo 20141**	Số lần HC:	**1**		
Mã số tài sản:	**-**	Mã số đơn:	**-**	Thời gian HC:	20		Phút
Ver chương trình:	**-**	Hạng mục yêu cầu:	**Đăng ký mới**	Nơi HC:	**Accretech**		
Mã số bản vẽ:	**-**						

Nhiệt độ	**Độ ẩm**	**Phán định**	**Người HC**	**Xác nhận**
20.0ºC	**50%**	**OK**	**Thắm 20427**	

Ghi chú:	ADKQHC của nhà cung cấp hiệu chuẩn

Kết quả hiệu chỉnh

THAM KHẢO KẾT QUẢ HIỆU CHỈNH ĐÍNH KÈM

BIỂU XÁC NHẬN KẾT QUẢ HIỆU CHỈNH NHÀ CUNG CẤP BÊN NGOÀI

Ngày xác nhận:		23/02/2026
Đảm trách	Xác nhận	Công nhân
Thắm 20427		

I. Kết quả xác nhận khi nhận máy

No	Nội dung xác nhận	Có	Không
1	Thiết bị nhận về có hư hỏng gì không		O
2	Thể thức thiết bị	O	
3	Giấy chứng nhận hiệu chỉnh	O	
4	Dấu công nhận tiêu chuẩn hiệu chỉnh	O	
5	Kết quả hiệu chỉnh	O	
6	Sơ đồ truy nguyên		O
7	Bản yêu cầu xác nhận đối ứng yêu cầu phòng thí nghiệm nội bộ theo tiêu chuẩn A16-2		O

II Xác nhận kết quả hiệu chỉnh

STT	Hạng mục hiệu chỉnh		Độ chính xác			
	Trị chuẩn Ra	Dung sai phán định	Kết quả hiệu chuẩn			
			Giới hạn dưới	Trị thực đo	Giới hạn trên	Phán định
1	0.41µm	± 5%	0.39µm	**0.409µm**	0.43µm	OK
			0.39µm	**0.411µm**	0.43µm	OK
			0.39µm	**0.411µm**	0.43µm	OK
			0.39µm	**0.409µm**	0.43µm	OK
			0.39µm	**0.412µm**	0.43µm	OK
2	3.19µm	± 5%	3.03µm	**3.187µm**	3.35µm	OK
			3.03µm	**3.191µm**	3.35µm	OK
			3.03µm	**3.197µm**	3.35µm	OK
			3.03µm	**3.197µm**	3.35µm	OK
			3.03µm	**3.188µm**	3.35µm	OK

Tiêu chuẩn phán định: Xem trong tài liệu S24-01

ACCRETECH

Control Number: TQR25362

Date of issue: December 8, 2025

Inspection report

TOKYO SEIMITSU CO., LTD.

Metrology Company

Quality Assurance Department

Customer: MABUCHI MOTOR DANANG LTD.

Object: Roughness specimen

Type: E-MC-S24D Serial number: 60994620LZ

Certified by:

Inspected by:

1. Inspection of *Ra* value of calibration surface

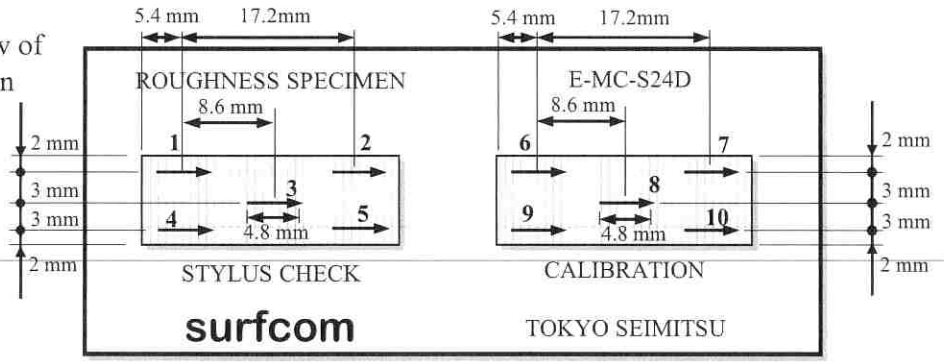
Judgement with the range of variation of *Ra* values on 5 positions.

Measuring Condition

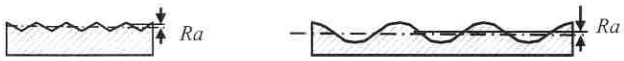
Stylus tip radius :	2.0 µm (79 µin)	Tilt correction	: Least squares mean line
Tracing speed :	0.3 mm/s (0.012 in/s)	Profile filter	: Gaussian
Evaluation length ln=5λc:	4.0 mm (0.16 in)	Cut-off wavelength λs	: 2.5 µm (98 µin)
Traversing length lt=ln+λc:	4.8 mm (0.19 in)	Cut-off wavelength λc	: 0.8 mm (0.031 in)

Stylus check surface	**Calibration surface**
Triangle wave length : 0.015 mm (591 µin)	Sinusoidal wave length : 0.1 mm (0.0039 in)

Top view of Specimen



Section view



Target surface	*Ra* of Stylus check surface			*Ra* of Calibration surface		
	Symbol	*Ra* (µm)	*Ra* (µin):reference	Symbol	*Ra* (µm)	*Ra* (µin):reference
Calibrated Value	**µ**	**0.410**	16.16	**µ**	**3.192**	125.67
Maximum value	Max	0.412	16.23	Max	3.197	125.88
Minimum value	Min	0.409	16.10	Min	3.187	125.46
Upper Variation	µ+5%	0.431	16.97	µ+5%	3.352	131.95
Lower Variation	µ-5%	0.390	15.35	µ-5%	3.032	119.39
Judgement	Conformance			Conformance		

End

Certificate Number : TQR25362

TQR25362
JCSS0047 MRA/IAJapan
2025-12

Certificate of Calibration

Customer	:	MABUCHI MOTOR DANANG LTD.
Address	:	LOTA2, NO.3 STREET, HOA KHANH INDUSTRIAL ZONE, HOA KHANH BAC WARD, LIEN CHIEU DISTRICT, DANANG CITY, VIETNAM
Object	:	Roughness specimen
Type	:	E-MC-S24D
Quantity	:	1
Serial number	:	60994620LZ
Manufacturer	:	Tokyo Seimitsu Co., Ltd.
Calibration Item	:	Surface texture: Roughness parameter *Ra* and *Rz*
Calibration Method	:	Roughness parameter measurement with stylus type surface texture instrument according to the laboratory calibration procedure DT78024-N001

Reference standard instruments used for calibration:

Instruments	Specifications	Control No.	Calibration date	Calibration Varidity	Calibration Lab.
Reference standard: Laser Interferometer	L-IM-21B	TQ2289	November 21, 2023	November 30, 2026	Tokyo Seimitsu JCSS0047
Suaface texture measuring instrument	SURFMASTER	TQ2286	May 19, 2025	May 31, 2026	Tokyo Seimitsu
Digital thermometer	LR8450-01	TQ2772	April 22, 2024	April 30, 2027	JEMIC JCSS0039

Calibration Site	:	4-2, Higashinakanuki-machi, Tsuchiura-shi, Ibaraki, Japan TOKYO SEIMITSU CO., LTD. Tsuchiura Plant Calibration room
Environmental Condition	:	Temperature : 20 °C±0.5 °C Relative humidity : 50 %±10 %
Date of Calibration	:	December 5, 2025

We certify that the result of calibration is in accordance with the following pages.

Date of Issue : December 8, 2025

4-2, Higashinakanuki-machi, Tsuchiura-shi, Ibaraki, Japan

TOKYO SEIMITSU CO., LTD.

Metrology Company, Calibration Laboratory

M. Takaku

Mr. Masakazu Takaku, General Manager

This certificate is based on article 144 of the Measurement Act and indicates the result of calibration in accordance with measurement standards traceable to Primary Measurement Standards (National Standards) which realizes the physical units of measurement according to the International System of Units (SI). The accreditation symbol is an attestation of which the result of calibration is treceable to Primary Measurement Standards (National Standards).

The certificate shall not be reproduced except in full, without the written approval of the issuing laboratory.

The calibration laboratory who issued this calibration certificate conforms to ISO/IEC 17025:2017.

This calibration certificate was issued by the calibration laboratory accredited by IAJapan who is a signatory to the Mutual Recognition Arrangement (MRA) of International Laboratory Accreditation Cooperation (ILAC) and Asia Pacific Accreditation Cooperation (APAC). This (These) calibration result(s) may be accepted internationally through ILAC/APAC MRA. Calibration label is to identify calibration status easily by attaching some information of Calibration Certificate to the calibrated item.

JCSS 0047

Certificate Number : TQR25362

Results of Calibration (JCSS calibration)

Object

Type	: E-MC-S24D	Serial number	: 60994620LZ

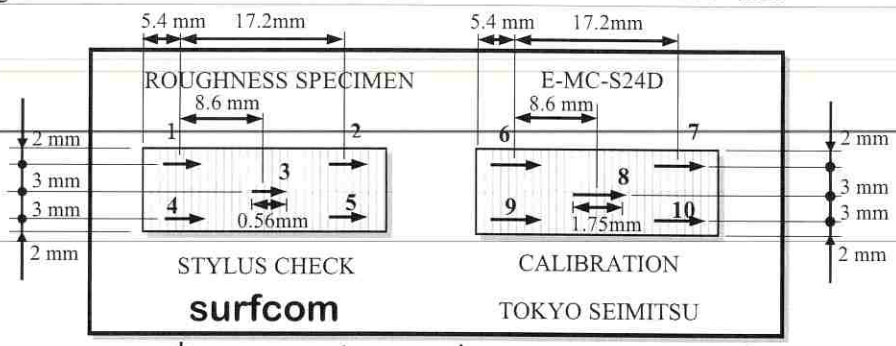
Calibration item : *Ra* and *Rz* according to ISO 4287 and ISO 4288

Measuring Condition

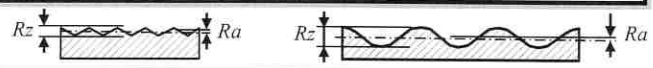
Stylus tip radius	: 2.0 μm	Measuring force	: 0.75 mN
Tracing speed	: 0.3 mm/s	Tilt correction	: Least squares mean line
Profile filter	: Gaussian	Cut-off wavelength λs	: 2.5 μm

	Stylus check surface(Small)	Calibration surface(Large)
Profile wavelength :	15 μm	100 μm
Cut-off wavelength λc :	0.08 mm	0.25 mm
Pre-travel (Weighting function length) *L* c:	0.08 mm	0.25 mm
Evaluation length ln=5λc :	0.40 mm	1.25 mm
Traversing length lt=ln+2*L* c :	0.56 mm	1.75 mm

Top view of specimen



Section view



Measuring position	*Ra, Rz* of stylus check surface			*Ra, Rz* of calibration surface		
	Symbol	*Ra* (μm)	*Rz* (μm)	Symbol	*Ra* (μm)	*Rz* (μm)
Left forward	1	0.406	1.48	6	3.145	9.90
Right forward	2	0.413	1.50	7	3.149	9.92
The center position	3	0.410	1.48	8	3.158	9.92
Left in front	4	0.407	1.48	9	3.151	9.92
Right in front	5	0.415	1.50	10	3.145	9.91
Number of measurement	*N*	5	5	*N*	5	5
Mean *Ra,Rz* = calibrated *Ra,Rz*	μ	**0.410**	**1.49**	μ	**3.150**	**9.91**
Standard deviation	σ	0.004	0.01	σ	0.005	0.01
Expanded Uncertainty	***U***	**0.018**	**0.19**	***U***	**0.019**	**0.19**

Note 1: **Calibrated *Ra* , *Rz*** are arithmetical mean of all measured *Ra* or *Rz* values on 5 positions.

Note 2: The standard deviation σ of all values on 5 positions.is included in the expanded uncertainties ***U***.

Note 3: The expanded uncertainty ***U*** (Coverage factor k = 2) are shown on above table.
The above reported expanded uncertainty of the calibration is stated as the standard uncertainty of calibration multiplied by the coverage factor k = 2 such that, for an approximately normal distribution, the coverage probability corresponds to approximately 95 %.

Note 4: The *Ra*'s on the table are calibrated on JCSS condition according to ISO 4287 and ISO 4288.
They are different from the "*Ra*" printed on the object for magnification adjustment with the former condition.

End.

Appendix with the value in inch

Results of Calibration (JCSS calibration)

Object

Type : E-MC-S24D Serial number : 60994620LZ

Calibration item : *Ra* and *Rz* according to ISO 4287 and ISO 4288

Measuring Condition

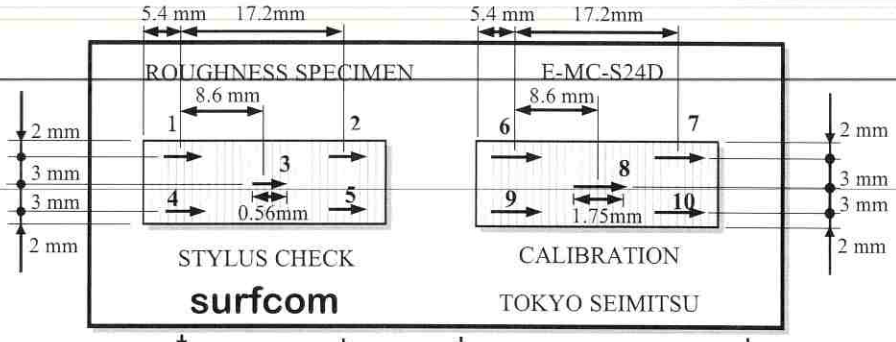
Stylus tip radius : 2.0 μm (79 μin) Measuring force : 0.75 mN

Tracing speed : 0.3 mm/s (0.012 in/s) Tilt correction : Least squares mean line

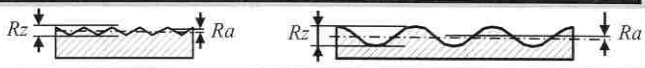
Profile filter : Gaussian Cut-off wavelength λs : 2.5 μm (98 μin)

	Stylus check surface(Small)	Calibration surface(Large)
Profile wavelength :	15 μm (591 μin)	100 μm (0.0039 in)
Cut-off wavelength λc :	0.08 mm (0.0031 in)	0.25 mm (0.0098 in)
Pre-travel (Weighting function length) *L* c:	0.08 mm (0.0031 in)	0.25 mm (0.0098 in)
Evaluation length ln=5λc :	0.40 mm (0.016 in)	1.25 mm (0.049 in)
Traversing length lt=ln+2*L* c :	0.56 mm (0.022 in)	1.75 mm (0.069 in)

Top view of specimen



Section view



Measuring position	*Ra, Rz* of stylus check surface					*Ra, Rz* of calibration surface				
	Symbol	*Ra* (μm)	***Ra*** (μin) :reference	*Rz* (μm)	***Rz*** (μin) :reference	Symbol	*Ra* (μm)	***Ra*** (μin) :reference	*Rz* (μm)	***Rz*** (μin) :reference
Left forward	1	0.406	15.97	1.48	58.2	6	3.145	123.81	9.90	389.7
Right forward	2	0.413	16.25	1.50	59.1	7	3.149	123.98	9.92	390.4
The center position	3	0.410	16.14	1.48	58.5	8	3.158	124.34	9.92	390.7
Left in front	4	0.407	16.02	1.48	58.2	9	3.151	124.06	9.92	390.6
Right in front	5	0.415	16.33	1.50	59.1	10	3.145	123.83	9.91	390.3
Number of measurement	*N*	5	5	5	5	*N*	5	5	5	5
Mean *Ra,Rz* = calibrated *Ra,Rz*	*μ*	**0.410**	16.14	**1.49**	58.6	*μ*	**3.150**	124.00	**9.91**	390.3
Standard deviation	σ	0.004	0.15	0.01	0.5	σ	0.005	0.22	0.01	0.4
Expanded Uncertainty	***U***	**0.018**	0.70	**0.19**	7.5	***U***	**0.019**	0.76	**0.19**	7.4

Note 1: **Calibrated *Ra* , *Rz*** are arithmetical mean of all measured *Ra* or *Rz* values on 5 positions.

Note 2: The standard deviation σ of all values on 5 positions.is included in the expanded uncertainties ***U***.

Note 3: The expanded uncertainty ***U*** (Coverage factor k = 2) are shown on above table.
The above reported expanded uncertainty of the calibration is stated as the standard uncertainty of calibration multiplied by the coverage factor k = 2 such that, for an approximately normal distribution, the coverage probability corresponds to approximately 95 %.

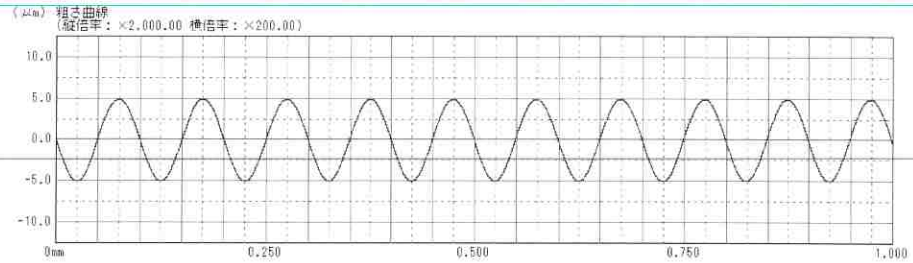
Note 4: The *Ra* 's on the table are calibrated on JCSS condition according to ISO 4287 and ISO 4288.
They are different from the "*Ra* " printed on the object for magnification adjustment with the former condition.

Appendix
Profile record on JCSS calibration

Type : E-MC-S24D Serial number: 60994620LZ

Measured Surface : Calibration surface (Large) Profile: Roughness profile

Recorded profile Note:The following figures show only the first portions of the evaluation lengths.

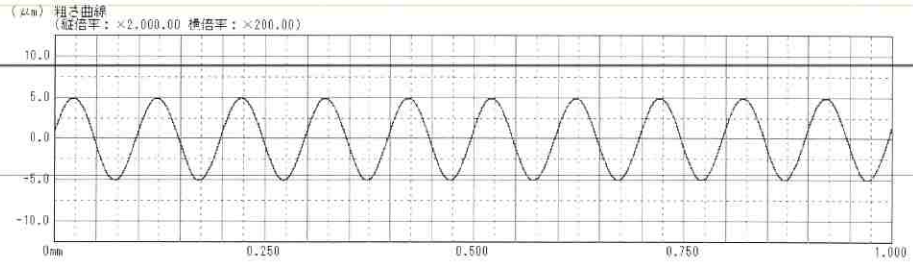


Position 6

Ra 3.145 μm

Rz 9.90 μm

RSm 99.8 μm

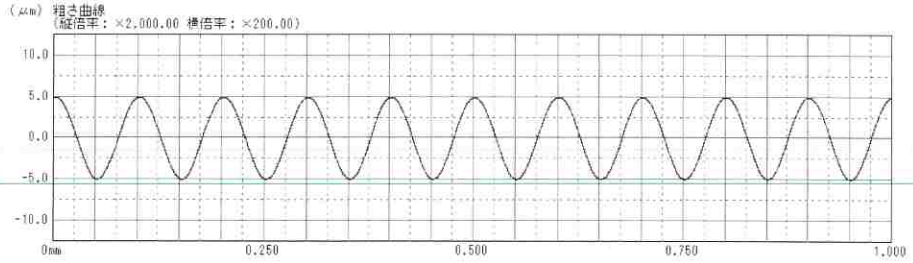


Position 7

Ra 3.149 μm

Rz 9.92 μm

RSm 99.8 μm

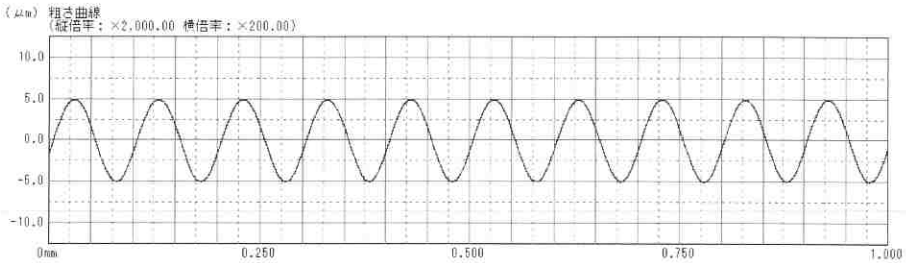


Position 8

Ra 3.158 μm

Rz 9.92 μm

RSm 99.8 μm

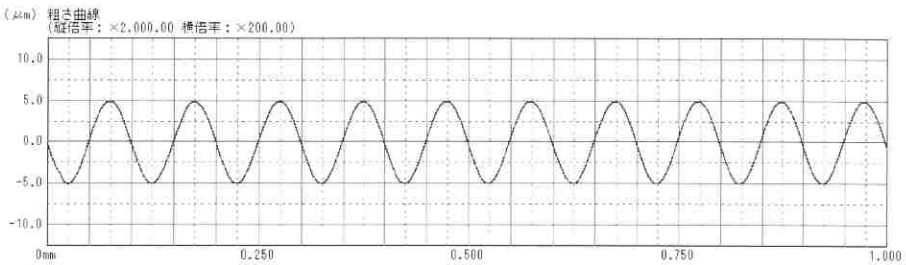


Position 9

Ra 3.151 μm

Rz 9.92 μm

RSm 99.8 μm



Position 10

Ra 3.145 μm

Rz 9.91 μm

RSm 99.8 μm

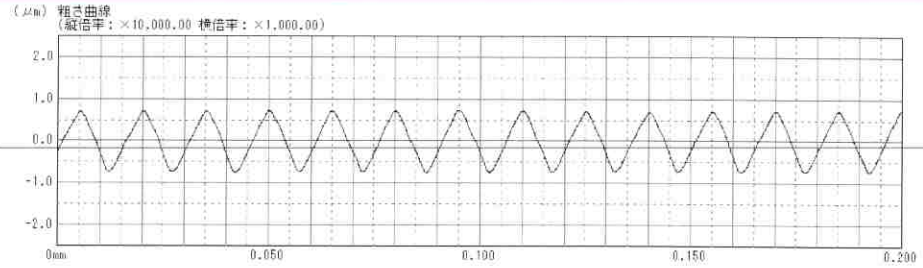
Appendix

Profile record on JCSS Stylus check surface

Type	: E-MC-S24D	Serial number:	60994620LZ
Measured Surface :	Stylus check surface (Small)	Profile:	Roughness profile

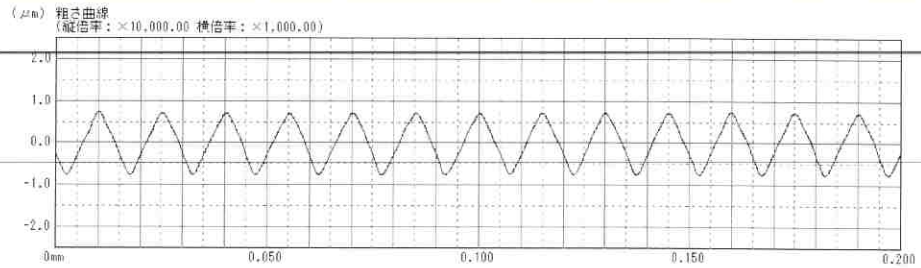
Recorded profile

Note:The following figures show only the first portions of the evaluation lengths.



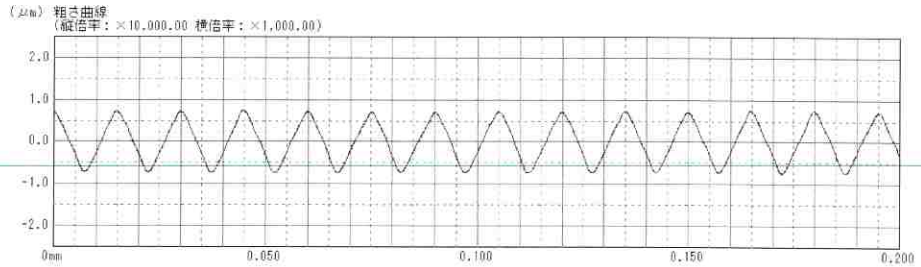
Position 1

Ra	0.406	μm
Rz	1.48	μm
RSm	15.0	μm



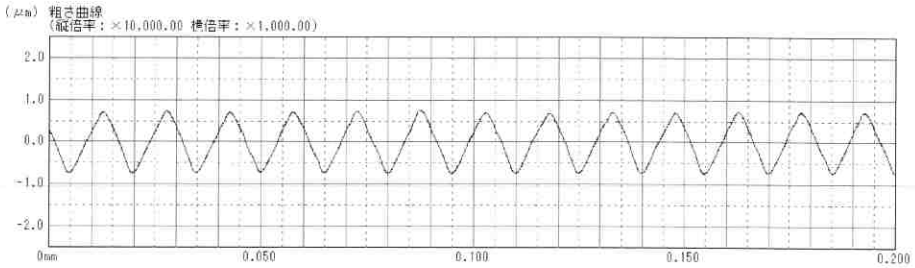
Position 2

Ra	0.413	μm
Rz	1.50	μm
RSm	15.0	μm



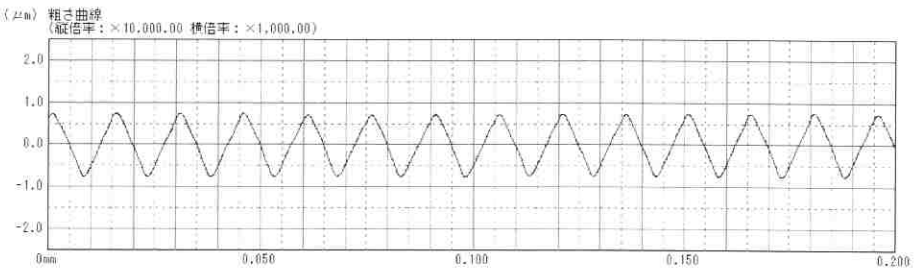
Position 3

Ra	0.410	μm
Rz	1.48	μm
RSm	15.0	μm



Position 4

Ra	0.407	μm
Rz	1.48	μm
RSm	15.0	μm



Position 5

Ra	0.415	μm
Rz	1.50	μm
RSm	15.0	μm

Appendix

Results of Calibration for Adjustment & Stylus check

Object

Type : E-MC-S24D Serial number : 60994620LZ

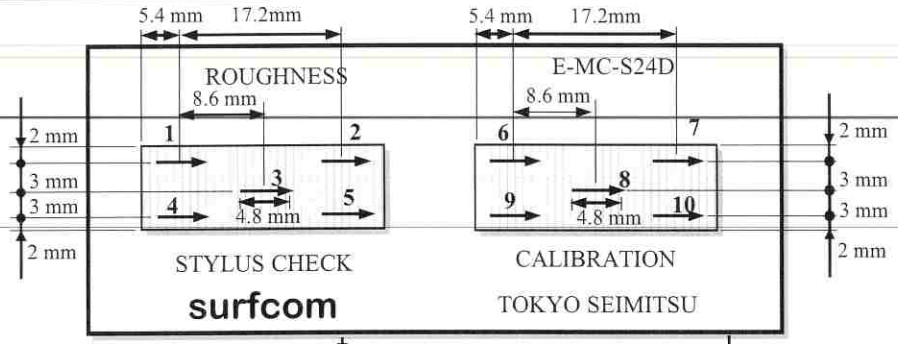
Calibration Item Arithmetical mean deviation of the roughness profile *Ra* (Indicated on the specimen)
But the applied conditions, λc= 0.8 mm and ln= 4 mm are different from ISO4288.

Measuring Condition

Stylus tip radius :	2.0 μm (79 μin)	Tilt correction	: Least squares mean line
Tracing speed :	0.3 mm/s (0.012 in/s)	Profile filter	: Gaussian
Evaluation length ln=5λc :	4.0 mm (0.16 in)	Cut-off wavelength λs :	2.5 μm (98 μin)
Traversing length lt=ln+λc :	4.8 mm (0.19 in)	Cut-off wavelength λc :	0.8 mm (0.031 in)

	Stylus check surface	Calibration surface
Profile wavelength :	15 μm (591 μin)	100 μm (0.0039 in)

Top view of Specimen



Section view

Measuring position	*Ra* of Stylus check surface			*Ra* Adjustment of Calibration surface		
	Symbol	*Ra* (μm)	*Ra* (μin):reference	Symbol	*Ra* (μm)	*Ra* (μin):reference
Left forward	**1**	0.409	16.10	**6**	3.187	125.46
Right forward	**2**	0.411	16.19	**7**	3.191	125.64
The center position	**3**	0.411	16.18	**8**	3.197	125.88
Left in front	**4**	0.409	16.10	**9**	3.197	125.85
Right in front	**5**	0.412	16.23	**10**	3.188	125.53
Number of measurement	***N***	5	5	***N***	5	5
Mean *Ra* = Calibrated *Ra* for adjust.	**μ**	**0.410**	16.16	**μ**	**3.192**	125.67
Standard deviation	σ	0.002	0.06	σ	0.005	0.19
Expanded Uncertainty	***U***	**0.016**	0.64	***U***	**0.019**	0.74

Note 1: **Calibrated *Ra*** for adjustment is arithmetical mean of all measured *Ra* values on 5 positions.

Note 2: The standard deviation sigma of all values on 5 positions is included in the expanded uncertainty ***U***.

Note 3: The expanded uncertainty ***U*** (coverage factor k = 2) are shown on above table.
The above reported expanded uncertainty of the calibration is stated as the standard uncertainty of calibration multiplied by the coverage factor k = 2 such that, for an approximately normal distribution, the coverage probability corresponds to approximately 95 %.

Note 4: As the condition for stylus check surface is not based on ISO4288 but company inspection standard, this page is without JCSS logo, and *Ra* on the table & specimen are only for stylus check.

Note 5: As the condition for caliblation surface is not based on ISO4288 but company inspection standard, this page is without JCSS logo, and *Ra* on the table & specimen are only for magnification adjustment.

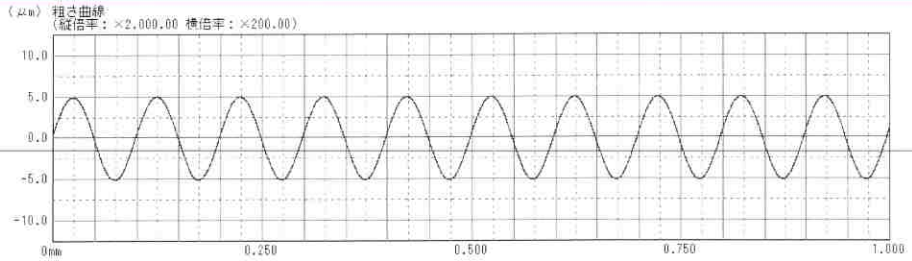
Appendix

Profile record on calibration surface (for adjustment)

Type : E-MC-S24D Serial number: 60994620LZ

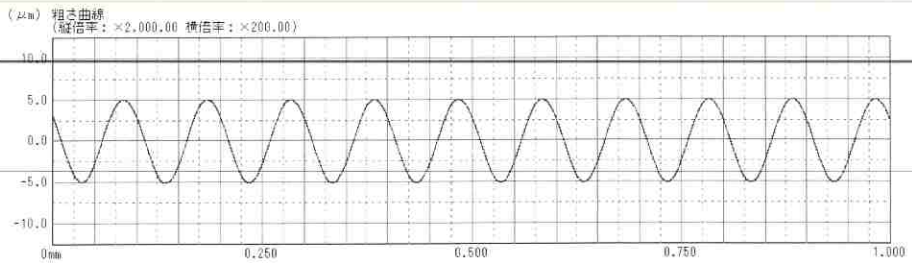
Measured Surface : Calibration surface (Large) Recorded profile: Roughness profile

Recorded profile Note:The following figures show only the first portions of the evaluation lengths.



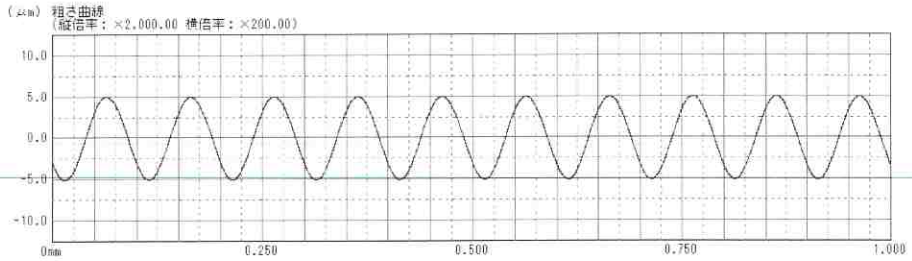
Position 6

Ra 3.187 μm



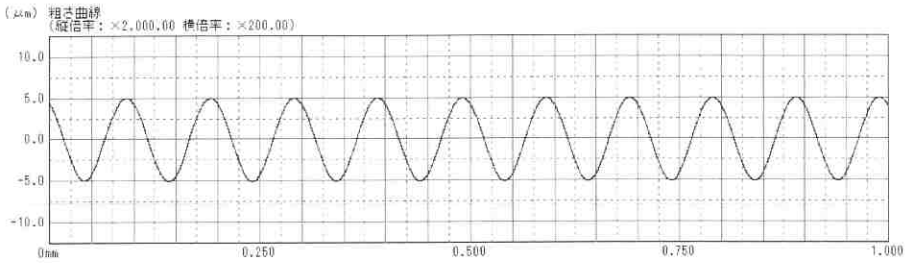
Position 7

Ra 3.191 μm



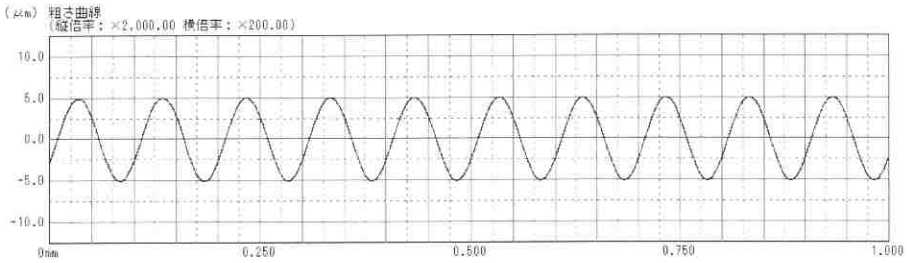
Position 8

Ra 3.197 μm



Position 9

Ra 3.197 μm



Position 10

Ra 3.188 μm

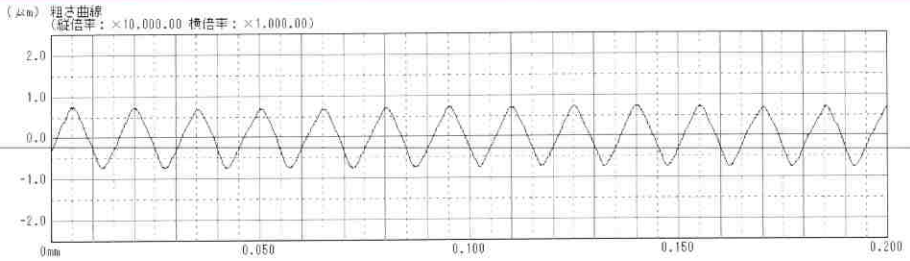
Appendix

Profile record on Stylus check surface

Type : E-MC-S24D Serial number: 60994620LZ

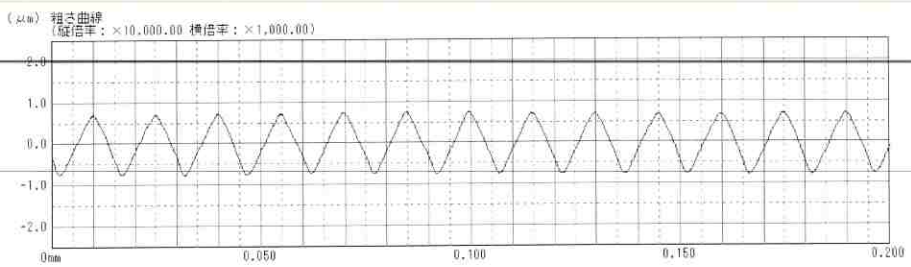
Measured Surface : Stylus check surface (Small) Recorded profile: Roughness profile

Recorded profile Note:The following figures show only the first portions of the evaluation lengths.



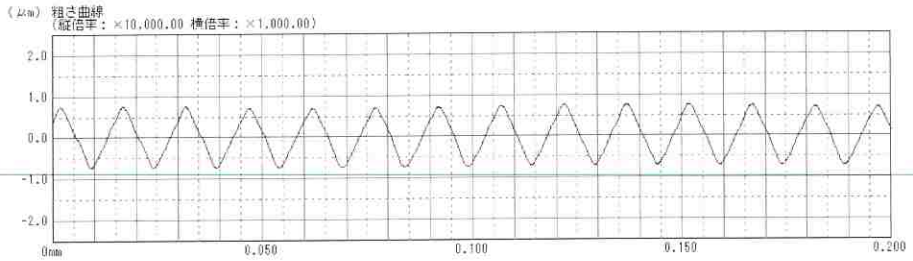
Position 1

Ra 0.409 μm



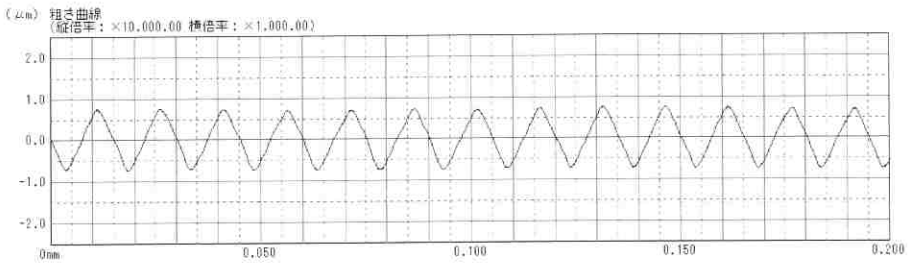
Position 2

Ra 0.411 μm



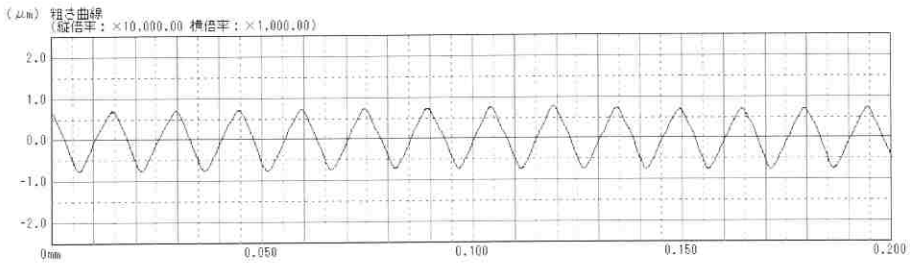
Position 3

Ra 0.411 μm



Position 4

Ra 0.409 μm



Position 5

Ra 0.412 μm

End of appendix

What is JCSS ?

Confident Evidence of Traceability and Competence!

JCSS

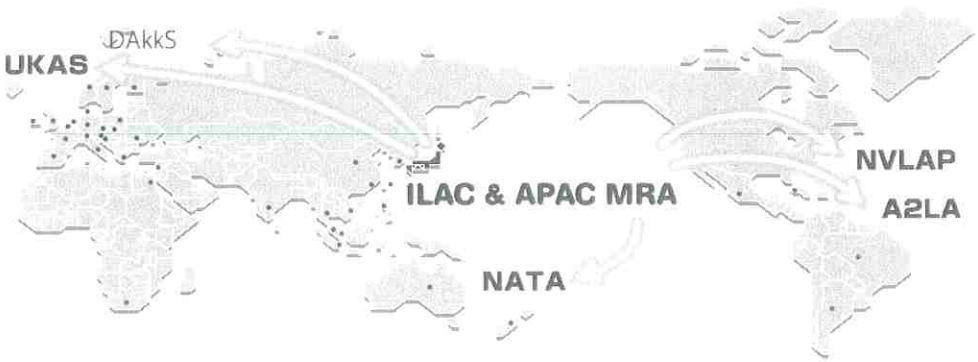
Japan Calibration Service System

JCSS is...

the Calibration Laboratory Accreditation System based on the Japanese Measurement Law in order to ensure the metrological traceability in Japan.
IAJapan, National Institute of Technology and Evaluation, Accreditation Body of JCSS assesses and approves the technical competence and conformity to the requirements of ISO/IEC17025 (JISQ17025) for calibration laboratories.

JCSS Calibration Certificate means...

JCSS Calibration Certificates issued by JCSS Laboratories, ***accredited by IAJapan under the ILAC and APAC MRA***, certify formally the metrological traceability to National Measurement Standards of Japan provided by ***NMIJ***.
There are no needs to trace back to upper traceability hierarchy if you are with a JCSS Calibration Certificate.
A JCSS Calibration Certificate is more confident "Evidence of Traceability" with Technical Competence and an appropriate Measurement Uncertainty than a Certificate with Legal metrology, QMS Certification, and Manufacturers' self declaration.
As IAJapan is one of signatories for ILAC and APAC MRA, JCSS calibration certificates are equivalent to those of laboratories accredited by other ILAC and APAC MRA signatories, such as NVLAP, A2LA, UKAS, DAkkS ***, NATA, etc.***
Also for calibration laboratories with overseas metrological traceability, IAJapan assesses them and grants accreditation under "ASNITE program."



(Example of JCSS Calibration Certificate)

For inquiries, please access to IAJapan

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